

Shadow Strategic Roadmap 2026-2028

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Version: 1.0 · **Prepared:** 2026-07-07 · **Post 4-agent full-web deep research Authors:** Alex Xiaoyu Ji (design + implementation) · Loredana C. Levitchi (regulatory-domain + banking-audience audit)

Purpose: consolidate the four parallel research reports (voice+AR UX, future direction, persona professionalization, RP1/Sony/Android XR/Apple design references) into one 18-month product + platform roadmap. Every claim below is cited to a dated public source in the appendix.

Executive summary

Shadow is on track to be the first shipping AR compliance product for regulated banking. Three simultaneous claims hold:

1. **Voice interaction pattern is settled:** gpt-realtime over WebRTC (200-350ms first-audio) as primary; Gemini 3.1 Flash Live as fallback for deictic references ("this loan"). No wake word — copy the visionOS 27 orb pattern shipping now.
 2. **Hardware bet is settled:** XREAL One Pro + Eye add-on (\$99) + iPhone/Mac tether + WebXR is the safest 24-month path. Apple killed Vision Pro 2 (Kuo confirmed 2026-06-03); Vision Pro is publicly de-prioritized. XREAL Aura Fall 2026 is upgrade path but not July demo path.
 3. **Three-vertical opportunity is real:** Bank (July 2026), Trading (Q4 2026 via Orallexa backend), Data-Science (Q1 2027). No shipping product spans all three; academic term "context-aware AR" (Zhu 2504.16562, Chen 2412.12681) not adopted by any vendor. Shadow claims "Ambient Council" as the surface layer term.
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Voice + AR interaction stack (July 2026 demo)

Primary voice model: gpt-realtime (OpenAI, released 2025-08). WebRTC transport, 200-350ms first-audio latency, native barge-in, MultiChallenge audio score 30.5% (up from 20.6% in prior generation). Reference: [OpenAI Aug 2025, webRTC Hacks latency benchmark](#).

Fallback voice model for deictic reference: Gemini 3.1 Flash Live. Best 1M-token multimodal context for "this loan" / "that flag" resolution across gaze + prior turns. Reference: [Google Blog](#).

Disqualified: Claude Voice (no real interruption, turn-by-turn only, cutoff bugs GH #40345 + #47136). Reference: [Anthropic Support docs](#).

Activation pattern: no wake word. Copy visionOS 27 Siri orb (June 2026 beta shipping now) — a gaze-anchored Council orb pinned above the physical loan PDF. Look-and-speak, 20-second always-listening

window after each utterance. Neural-Band-style pinch as push-to-talk fallback for noisy demo rooms.

References: [Road to VR visionOS 27](#), [MacRumors 2026-06-09](#).

Interaction contract: ambient conversation with native barge-in (“wait, back up”). Every AI answer anchored as visual overlay attached to the specific voice or flag being discussed per arXiv 2405.18537 (CHI 2024).

Compliance layer: persist voice queries + transcripts into Shadow’s hash-chain attestation (v1.5.10 primitive already exists). Treat every voice query as evidentiary. Disclose “you are speaking to an AI compliance council” per TX TRAIGA (Jan 2026) + FCC 2024 declaratory ruling at session start.

Hardware bet — 18-month view

Shipped or shipping in the demo window (July 2026)

- **XREAL One Pro** (\$599) — Lora purchased 2026-07-06. Display peripheral, requires USB-C DP-Alt from Android or Mac.
- **XREAL Eye add-on** (\$99) — 12MP camera + 6DoF spatial anchoring. GA since July 2025. Order immediately.
- **Meta Ray-Ban Meta Gen 2** (\$379) — only mass-market smart-glasses installed base. Audio-only mode as fallback demo path if XREAL fails.
- **Apple Vision Pro M5** (\$3,499) — shipped Oct 2025 but publicly de-prioritized (Kuo 2026-06-03). Skip for the pitch; too expensive for banking narrative.
- **Samsung Galaxy XR** (\$1,799) — shipped Oct 2025, Android XR + Gemini baked in. Path forward if XREAL Aura slips. Reference: [Road to VR](#).

Coming in the roadmap window (Fall 2026 - 2028)

- **XREAL Aura** (\$1,500, Fall 2026) — standalone Android XR + Gemini + puck. Same WebXR + Chrome path as our current stack; migration cost near zero. Frame in pitch as upgrade path.
- **Snap Specs consumer** (\$2,195, Fall 2026) — 51° FOV standalone AR + OpenAI + Google AI. Prosumer/dev.
- **Meta Hypernova / Artemis** (~\$800 Sept 2026 / 2027) — Ray-Ban form + sEMG wristband. Consumer mass market. Watch, don’t build for.
- **Apple smart glasses** (2027, Kuo) — audio-only v1, display glasses “2028 or later.” Not on Shadow’s critical path.

Disqualified

- **Meta Quest 3S** (\$299) — VR headset with passthrough. Breaks eyeglass aesthetic; not a compliance officer sitting-at-desk device.
- **OpenAI/Ive smart speaker** (Feb 2027) — not glasses at launch.
- **xAI/Grok glasses** — vaporware, Musk denied July 2026.

Three-vertical roadmap

Bank Mode (July 2026 demo, Q4 2026 GTM)

- **Product:** Shadow v3 Ambient Council on XREAL One Pro + Eye + iPhone.
- **Audience:** Chief Compliance Officer + Chief Audit Executive at mid-tier banks (Raymond James, Stifel, LPL, Houlihan Lokey, Lazard MM). Not JPM/GS (they build in-house).
- **Pricing:** \$2K/seat/year. Comparable: Vanta \$15-60K/year, Drata \$15-75K/year, enterprise SOC 2 \$600K+.
- **2027 realistic TAM:** 30 mid-tier banks × 20 seats × \$2K = **\$1.2M ACV** path to \$5M by 2028 if Big-4 channel partnership lands (Deloitte Omnia serves 85K auditors; PwC Aura; KPMG Clara). References: [Deloitte Omnia](#), [KPMG Clara](#), [Pento 2025 review](#), [ItBudgetCalculator](#).
- **Regulatory anchor:** SR 26-2 Tier 3 explicitly carves GenAI/agentic AI out — Shadow fills the gap.

Trading Mode (Q4 2026 — Orallexa backend)

- **Product:** Shadow surface + Orallexa backend (github.com/alex-jb/orallexa-ai-trading-agent , LangGraph 5-voice Bull/Bear/Judge + Critic + Polyseer, 745+ tests).
- **Audience:** Chief Risk Officer, not the trader. Compliance sidecar on the desk, not Bloomberg-seat replacement.
- **Anchor:** Bloomberg / broker screen. Risk-Officer voice surfaces VaR/ES violation pill anchored to position line; AML/KYC voice surfaces tipping-off flags anchored to counterparty row.
- **Direct competitor:** [Pillar \(\\$20M a16z seed Apr 2026\)](#) — but they don't have AR. Shadow's differentiator is the surface layer + attestation chain.

Data-Science Mode (Q1 2027)

- **Product:** Shadow surface + calibration engine on Jupyter cell / Cursor buffer anchor. Fair Lending voice flags protected-class proxy features inline (per v1.5.9 32 adversarial tests).
- **Audience:** ML validator, model validation officer. Quiet but defensible — SR 26-2 Tier 3 explicitly carves out GenAI, no incumbent owns "council-based validation."
- **Reference customer:** Loredana's Columbia statistics faculty introduction (2026-07-06 promise); Prof. Yang's capstone cohort.

Persona professionalization (v1.5.14, shipping now)

Post-deep-research on CFPB Circular 2022-03/2023-03 + FFIEC ECOA procedures + SR 26-2 replacement letter + FinCEN CDD Rule + tipping-off provisions, all 5 core persona rationale strings updated with real regulatory examiner voice.

Before/After sample (Credit Fundamentals FICO block path): - **Before:** "FICO=X below Addendum A floor Y; credit-eligibility floor failure is a hard block per Levitchi 2026-06-19 policy semantics." - **After:** "FICO=X below Addendum A floor Y (Reg B §1002.6 permissible consideration); credit-eligibility floor

failure is a hard block per Levitchi 2026-06-19 policy semantics. Principal reason per CFPB Circular 2022-03: insufficient credit history depth (AA01)."

Before/After sample (Risk Officer): - **Before:** "VaR(95%, 10d)=X% vs 12% Addendum C cutoff (status: escalate); LTV=0.82 vs 0.80 Addendum C ceiling." - **After:** "SR 26-2 materiality assessment: VaR(95%, 10d)=X% vs 12% Addendum C benchmark (status: escalate); LTV=0.82 vs 0.80 Addendum C ceiling. Effective challenge: exposure x purpose warrants escalation to Credit Committee."

Before/After sample (Fair Lending block): - **Before:** "Fair-lending review flag set; ECOA/Reg B disparate-impact escalation required prior to human review." - **After:** "Fair-lending review flag set; FFIEC three-step framework required — prima facie statistical showing, business necessity defense, then less discriminatory alternative search per ECOA/Reg B §1002.4. Escalate for comparative file review prior to human decision."

All 5 personas updated. 706/707 tests pass, zero regressions.

Design references (Ready Player One + Sony + Android XR + Apple)

Adopt from RP1: Kinetic HUD summon on button/context trigger + dissolve on decision. Warm/cold faction color language (approve = warm, block = cold). References: [Territory Studio](#), [fxguide](#).

Adopt from Sony: 360 Reality Audio object-based spatial-audio primitive. 5 personas as 5 spatial audio object sources in a 145° arc around reviewer. References: [Sony 360 RA overview](#).

Adopt from Android XR: WebXR delivery. Same build runs Galaxy XR + Aura + Vision Pro Safari + Chrome desktop. References: [Android Developers WebXR](#), [XREAL Aura announcement](#).

Adopt from Apple Vision Pro: Environment-anchor everything, never head-lock. Gaze-hover + pinch-select for XREAL Eye users. Windows anchored to space, not head. References: [Apple HIG for visionOS](#), [Ken Pfeuffer gaze-pinch](#).

Reject: Vision Pro Persona avatars (uncanny + breaks Severance restraint), Ready Player One haptic suit (fiction), Sony XYN ring controller (separate hardware), sticky head-locked overlay (Apple HIG anti-pattern), Gemini auto-spatialize on demo path (Android-only cuts Apple auditors out).

18-month execution plan

Quarter	Deliverable	Success criterion
2026-Q3 (July)	v3 Ambient Council Bank mode demo to Yeshiva Dean + VP	Live 5-question voice interaction on XREAL One Pro + Eye + iPhone, hash-chain attestation verified on-stage
2026-Q3 (Aug)	ICAIF Milan paper submit (already skeleton'd 2026-07-06)	Paper accepted, Yang + Levitchi co-authors
2026-Q3 (Aug)	IEEE VR 2027 abstract submit (v3 abstract already drafted)	Abstract accepted for full paper track
2026-Q4	Trading Mode v3.1 — Orallexa backend + Bloomberg-screen anchor	Working demo on 3 SpaceX-IPO-Tracker tickers, mid-tier trading desk pilot conversation
2027-Q1	Data-Science Mode v3.2 — Jupyter + Cursor anchor	Working demo with Columbia statistics faculty + Yang capstone cohort
2027-Q1	Big-4 channel conversation (Deloitte / PwC / KPMG / EY)	Warm intro via Loredana or Yang's network, LOI or pilot conversation
2027-Q2	Audit Replay v3.3 — Joi-style volumetric persona embodiment	Ships on XREAL Aura (standalone). First 3 bank counsel evaluations.
2027-Q3	Mid-tier bank first customer contract	1 bank at \$40K ACV (20 seats × \$2K), 12-month contract
2028-Q1	Path to \$1M ARR	5 mid-tier banks × \$200K average ACV, 100+ compliance officer seats

Contact + attribution

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This roadmap is a public artifact of `shadow-mentor` (MIT). Every claim above is cited to a dated public source. Open an issue if any claim is out of date.